D.K.T.E. Society's Textile and Engineering Institute, Ichalkaranji.

(An Autonomous Institute, Affiliated to Shivaji University, Kolhapur)

Accredited with 'A+' Grade by NAAC

Department of CSE (AIML) Engineering

2025-2026

Promoting Excellence in
Teaching, Learning & Research

Internship Training

Web Programming with python

Submitted By:

23UAM076	Gouri Hemant Mangore
23UAM071	Sakshi Ajit Magdum
23UAM053	Rasika Sadashiv khade

INTERNSHIP CERTIFICATES

SUNBEAM

CERTIFICATE

This is to certify that

Gouri Hemant Mangore

Student from AI-ML of DKTE Engg Institute, Ichalkaranji

has successfully completed the

Hands-on Training and Internship Programme in Web Programming with Python

during the period

10-Dec-2025 To 03-Jan-2026

and has been evaluated & awarded a Grade of "A"

Nilesh Ghule
Technical Director

Nitin Kudale
Managing Director

SunBeam Infotech Pvt. Ltd., Pune

SUNBEAM

CERTIFICATE

This is to certify that

Sakshi Ajit Magdum

Student from AI-ML of DKTE Engg Institute, Ichalkaranji

has successfully completed the

Hands-on Training and Internship Programme in Web Programming with Python

during the period

10-Dec-2025 To 03-Jan-2026

and has been evaluated & awarded a Grade of **"A"**

Nilesh Ghule
Technical Director

Nitin Kudale
Managing Director

SunBeam Infotech Pvt. Ltd., Pune

SUNBEAM

CERTIFICATE

This is to certify that

Rasika Sadashiv Khade

Student from AI-ML of DKTE Engg Institute, Ichalkaranji

has successfully completed the

Hands-on Training and Internship Programme in Web Programming with Python

during the period

10-Dec-2025 To 03-Jan-2026

and has been evaluated & awarded a Grade of "B"

Nilesh Ghule
Technical Director

Nitin Kudale
Managing Director

SunBeam Infotech Pvt. Ltd., Pune

Index

Table of content Page No.

1. Introduction 6
2. Problem Statement 7
3. Objectives 8
4. Technology Stack 9
5. System Architecture 12
6. Database Design 14
7. Working Methodology 15
8. About the Projects 16
9. Output provided to the End Client 16
10. Benefits to the Client 16
11. Our roles 17
12. Conclusion 18

7

1. Introduction

The Sunbeam Online Student Portal is a full-stack web application designed to provide a centralized platform for online learning and course management. The system allows students to register, log in, access courses, and watch educational videos, while the admin can manage students, courses, and video content efficiently.

The application is developed using ReactJS for the frontend, Flask for the backend, and MySQL for the database. It ensures secure access using JWT authentication and role-based access control, allowing different functionalities for Admin and Student users.

This project helps in simplifying the learning process by providing structured content and secure user management in a single platform.

2. Problem statement

There is no centralized and secure system for managing online courses and student data efficiently. Students face difficulty in accessing structured learning content, and administrators struggle to manage courses and users in an organized way.

3.Objectives:

The main objectives of this project are:

- To develop a user-friendly online course portal
- To provide detailed information about available courses
- To enable easy and quick course registration
- To design a responsive interface accessible across devices
- To implement efficient data storage and retrieval mechanisms
- To demonstrate the use of modern web technologies in real-world applications

4. Technology Stack :

4.1. ReactJS

ReactJS is used for building the frontend user interface. It helps in creating reusable components and provides a smooth user experience.

4.2. Flask

Flask is used as the backend framework to develop RESTful APIs and handle business logic.

4.3. MySQL

MySQL is Used as the relational database to store user data, courses, and videos in a structured format.

4.4. JWT (JSON Web Token)

JWT is used for secure authentication and authorization, ensuring only authorized users can access the system

4.5. Postman

Postman is used for testing APIs and verifying backend functionality.

4.6. VS Code

VS Code is used as the development environment for writing and managing code.

4.7.Libraries:

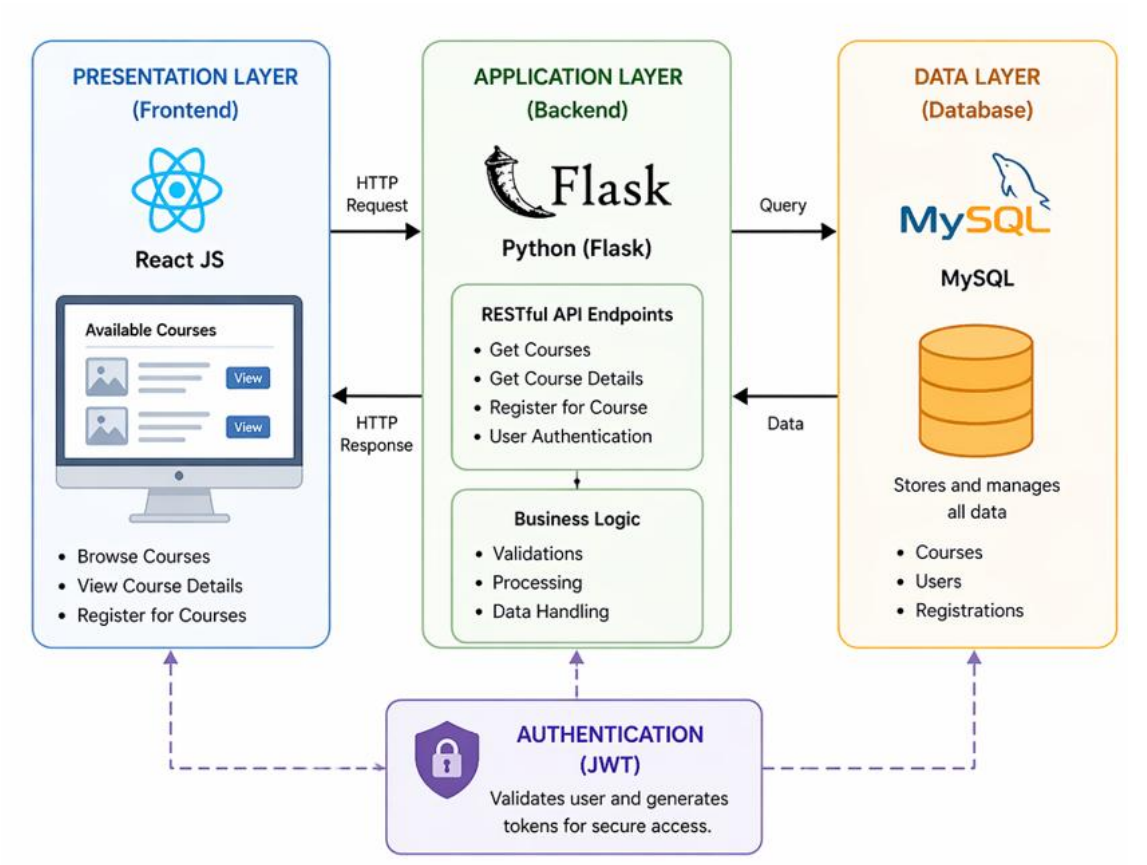
Backend (Flask - Python)

- **Flask** – Used to build the web application and create REST APIs
- **Flask-JWT-Extended** – Used for implementing JWT authentication and securing APIs
- **Flask-CORS** – Used to handle cross-origin requests between frontend and backend
- **mysql-connector-python** – Used to connect and interact with the MySQL database

Frontend (ReactJS)

- **React Router** – Used for navigation between different pages
- **Axios / Fetch API** – Used to send HTTP requests to backend APIs
- **Bootstrap** – Used for styling and responsive UI design

5.System Architecture:



The Sunbeam Online Course Portal follows a three-tier architecture, which separates the application into three main layers: frontend, backend, and database. This architecture ensures better scalability, maintainability, and performance of the system.

1. Presentation Layer (Frontend)

The presentation layer is developed using React JS, which provides a dynamic and responsive user interface. It allows users to interact with the system by browsing courses, viewing details, and registering for courses. This layer communicates with the backend through API requests.

2. Application Layer (Backend)

The application layer is implemented using Python (Flask). It handles all business logic, processes user requests, and manages communication between the frontend and the database. It also includes API endpoints that are used for operations like fetching course data and handling user registrations.

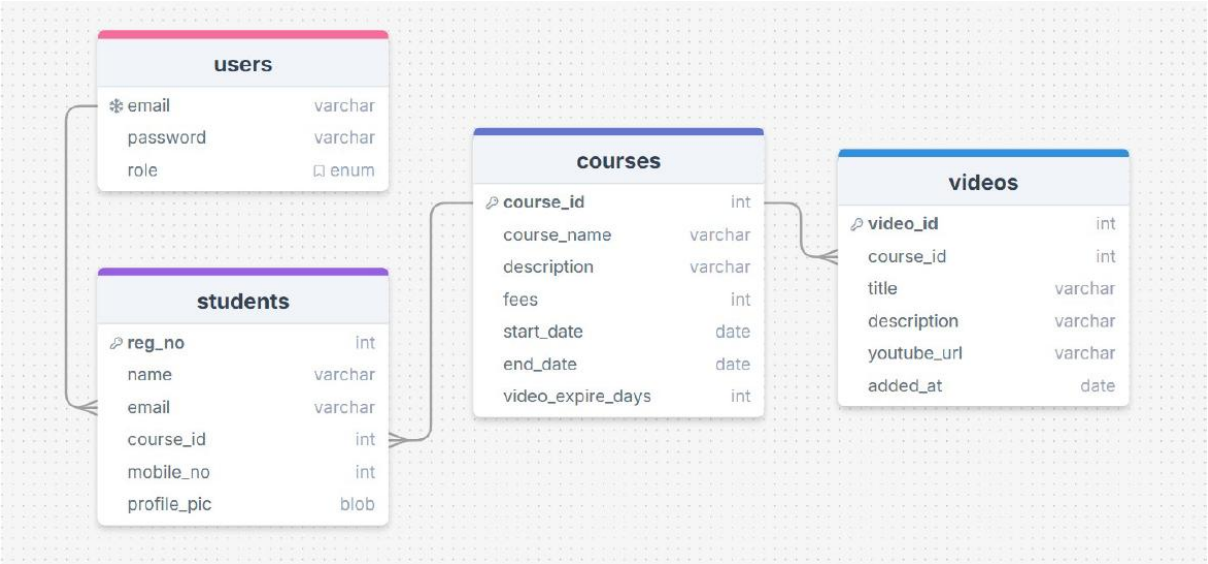
3. Data Layer (Database)

The data layer uses MySQL to store and manage all the application data, including course details and user information. It ensures data consistency, security, and efficient retrieval of information.

***Authentication Mechanism**

The system uses JWT (JSON Web Token) for secure authentication. It ensures that only authorized users can access certain features of the application by validating tokens during each request.

6.Database Design:



The database is designed using a relational model with four main tables: Users, Students, Courses, and Videos. The Users table handles authentication and role management, while the Students table stores student details. The Courses table contains course information, and the Videos table stores videos linked to each course.

Relationships between tables are maintained using primary and foreign keys, ensuring structured data storage and efficient data retrieval.

7. Working methodology

The Sunbeam Online Student Portal follows a structured workflow based on client-server architecture. The working of the system is divided into the following steps:

Step 1: User Authentication

Users (Student/Admin) log in using their credentials. The backend verifies the data and generates a JWT token for secure access.

Step 2: Role-Based Access

Based on the user role (Admin or Student), access is granted to specific functionalities using role-based authorization.

Step 3: API Communication

The frontend (ReactJS) communicates with the backend (Flask) through REST APIs to send and receive data.

Step 4: Data Processing

The backend processes requests, applies business logic, and interacts with the MySQL database to fetch or store data.

Step 5: Student Functionality

Students can view available courses and watch videos after successful authentication.

Step 6: Admin Functionality

Admin can add, update, and delete courses, manage videos, and view student data.

Step 7: Secure Data Flow

All requests are validated using JWT tokens to ensure secure and authorized access.

8. About the Project:

The Sunbeam Online Student Portal is a web-based application designed to provide a structured and secure platform for online learning. The system allows students to register, log in, and access course content in the form of videos.

The application includes two main modules: Student and Admin. The Student module enables users to view courses and watch videos, while the Admin module provides functionalities to manage students, courses, and video content.

The system is built using ReactJS for the frontend, Flask for the backend, and MySQL for database management. Communication between frontend and backend is handled through REST APIs, and security is maintained using JWT authentication and role-based access control.

9. Output provided to the End Client :

The system provides a user-friendly platform where students can securely log in, access available courses, and watch educational videos in a structured manner. It ensures smooth navigation and organized content delivery.

The admin is provided with a dashboard to manage students, courses, and videos efficiently, including adding, updating, and deleting data.

The system also ensures secure access using JWT authentication, preventing unauthorized usage. It provides role-based access so that each user can only perform allowed actions.

Overall, the application delivers a centralized and efficient solution for managing and accessing online learning resources.

10.Benefits to the Client

The system provides a centralized platform for managing and accessing educational content, reducing manual effort and improving organization.

It ensures secure access using JWT authentication and role-based control, preventing unauthorized usage.

Students benefit from easy access to structured courses and video content anytime, improving learning convenience.

Admins can efficiently manage students, courses, and videos through a single dashboard, saving time and effort.

The system improves overall efficiency, data management, and user experience in online learning.

11. Our Roles:

Gouri Hemant Mangore – Backend Developer

Responsible for backend development using Flask. Developed RESTful APIs for user authentication, course management, and video handling. Implemented JWT-based authentication and role-based access control to ensure secure system access.

Sakshi Ajit Magdum – Database & Testing Engineer

Responsible for designing the MySQL database, creating tables, and maintaining relationships between users, courses, and videos. Performed API testing using Postman and ensured proper functionality and error handling.

Rasika Sadashiv Khade – Frontend Developer

Responsible for developing the user interface using ReactJS. Designed Student and Admin modules including login, registration, dashboard, course list, and video pages. Integrated frontend with backend APIs for smooth data interaction.

12. Conclusion:

The Sunbeam Online Student Portal is a full-stack web application that provides a secure and organized platform for online learning and data management. It successfully implements JWT-based authentication and role-based access control to ensure secure and restricted access for different users.

The system allows students to easily access courses and videos, while the admin can efficiently manage all data through a centralized dashboard.

This project helped in understanding full-stack development, API integration, database management, and security concepts like authentication and authorization.